

```
from google.colab import files
uploaded = files.upload()
```

Screenshot ... 142020.png

Screenshot 2026-07-13 142020.png(image/png) - 268920 bytes, last modified: 13/07/2026 - 100% done
Saving Screenshot 2026-07-13 142020.png to Screenshot 2026-07-13 142020 (1).png

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

# Use one of the uploaded image paths
image_path = 'Screenshot 2026-07-13 142020 (1).png'

img = cv2.imread(image_path)

if img is None:
    print(f"Error: Could not load image from {image_path}. Please make sure the file exists and is ac
else:
    rows, cols = img.shape[:2]

    # Define source and destination points for perspective transformation
    src_points = np.float32([[0, 0], [cols - 1, 0], [0, rows - 1], [cols - 1, rows - 1]])
    # These destination points create a trapezoidal distortion
    dst_points = np.float32([[0, 0], [cols - 1, 0], [int(0.33 * cols), rows - 1], [int(0.66 * cols),

    # Get the perspective transformation matrix
    M = cv2.getPerspectiveTransform(src_points, dst_points)

    # Apply the perspective transformation
    perspective_img = cv2.warpPerspective(img, M, (cols, rows))

    # Convert images from BGR to RGB for correct display with matplotlib
    img_rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
    perspective_img_rgb = cv2.cvtColor(perspective_img, cv2.COLOR_BGR2RGB)
```

```
# Display the original and transformed images using matplotlib
plt.figure(figsize=(12, 6))

plt.subplot(1, 2, 1)
plt.imshow(img_rgb)
plt.title('Original Image')
plt.axis('off')

plt.subplot(1, 2, 2)
plt.imshow(perspective_img_rgb)
plt.title('Perspective Transformed Image')
plt.axis('off')

plt.show()

# If you still want to save the transformed image to disk, uncomment the line below:
# cv2.imwrite('Perspective_Transformed_Image.jpg', perspective_img)
```

Original Image

Successfully downloaded input.jpg



Cause of Aliasing:

- Aliasing occurs when the image is sampled below the Nyquist rate.
- High-frequency details appear as jagged edges or moiré patterns.
- Quantization reduces pixel intensity precision but is not the main cause of

Corrective Approach:

- Apply an anti-aliasing (Gaussian) filter before downsampling.
- Use better interpolation methods such as INTER_LINEAR or INTER_AREA.
- Increase the sampling resolution if possible.

Perspective Transformed Image

Successfully downloaded input.jpg



